



CS & IT ENGINEERING



Algorithms

Analysis of Algorithms

Lecture No.- 05



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

$0, \Omega, \theta$
Practice
Mathematical Properties

Topics to be Covered



Topic

Topic

Topic

Properties

Small Notations



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Analysis of Algorithms

1. Big Oh (UB)

$$f(n) = O(g(n))$$

Iff $f(n) \leq c * g(n)$, some $c > 0, n \geq n_0$

2. Big Omega (Ω)

$$f(n) = \Omega(g(n))$$

Iff $f(n) \geq c * g(n)$, some $c > 0, n \geq n_0$

3. $F(n) = \Theta(g(n))$

$$f(n) = O(g(n))$$

and

$$F(n) = \Omega(g(n))$$



Topic : Analysis of Algorithms

#Q. $f(n) = \sum_{i=1}^n \log(i)$

$\theta(n \log n)$

A

$F(n) = O(\log n)$

B

$F(n) = O(\log n)$

C

$F(n) = \Omega(\log n)$

D

$F(n) = O(\log n)$



Topic : Analysis of Algorithms

Sol. $f(n) = \sum_{i=1}^n [\log(i)]$

$$= \log(1) + \log(2) + \log(3) + \dots + \log(n)$$

$$= \log(1 * 2 * 3 * \dots * n)$$

$$f(n) = \log(n!).$$



Topic : Analysis of Algorithms

Method:- 1.

Logical.

$$\text{Log}(n) \leq \log(n)$$

$$\log(n) + \log(n+1) \leq \log(n) + \log(n)$$

$$\log(n) + \log(n-1) \dots \log(1) \leq \log(n) + \log(n) \dots + \log(n) \rightarrow \text{n times}$$

$$\log(n) + \dots \log(1) \leq n * \log(n)$$

$$\log(n!) \leq n * \log(n) \Rightarrow \log(n!) = O(n \log n)$$



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Method:- 2.

Stirling's

Using string's Approximation



$$n! \approx \sqrt{2\pi n} * \left(\frac{n}{e}\right)^n$$



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$$\begin{aligned}f(n) &= \sum_{i=1}^n \log(i) = \log(n!) \\&= \log\left(\sqrt{2\pi n} * \left(\frac{n}{e}\right)^n\right) \rightarrow \text{using stirting's Approx.} \\&= \log(\sqrt{2\pi}) + \frac{1}{2}\log(n) + n[\log(n) - \log(e)] \\&= \left[\log(\sqrt{2\pi}) + \frac{1}{2}\log(n) + n * \log(n) - n * \log(e)\right]\end{aligned}$$

$$\begin{aligned}f(n) &= O(\underline{n \log n}) \\f(n) &= \Omega(\underline{n \log n})\end{aligned} \rightarrow \log(n!) = \underline{\underline{\theta(n \log n)}}$$



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Bonus logic → Method -3 → Using Integration

$$f(n) = \sum_{i=1}^n \log(i) = \log(1) + \log(2) + \dots + \log(n)$$

$$= \int_1^n \log(x) dx = [x(\log x - 1)]_1^n$$

$$= n \log n - n + c$$

$$= \cancel{O(\log n)}$$

$$= \underline{O(n \log n)}$$



Topic : Analysis of Algorithms

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#Q. $f(n) = \sum_{i=1}^n i^3 = x$ choice for x .

✓ I. $\theta(n^4)$

✗ II. $\theta(n^5)$

✓ III. $O(n^5)$

✓ IV. $\Omega(n^3)$

59%

A I, II, III ✗

B II, III, IV ✗

C I, II, III, IV ✗

✓ **D** I, III, IV

D

$$\text{Sol: } \sum_{i=1}^n i^3 = 1^3 + 2^3 + \dots + n^3$$

$$= \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^2(n+1)^2}{4} = \frac{n^2(n^2+2n+1)}{4}$$

$$= \left[\frac{n^4 + 2n^3 + n^2}{4} \right] = O(n^4)$$

$O(n^4)$ $\uparrow$
 $\searrow$ $\Omega(n^4)$ $\downarrow$
 $= \Omega(n^5)?$ $\times$
 $= O(n^5)$ $\checkmark$ $\neq O(n^5)$

$\Omega(n^3)$ $\checkmark$



Topic : Analysis of Algorithms

The big notation (O, Ω) provides the upper bounded & Lower Bound that may or may not be tight.

~~Tight LB:~~

tight

(Can be as well as loose bounded)

E.g.1. $\boxed{\Omega(n)} \rightarrow \boxed{n+10} \rightarrow \boxed{O(n)} \rightarrow \text{Tight UB}$

Tight LB $\leftarrow$

$\left. \begin{array}{l} \Omega(\sqrt{n}) \\ \Omega(1) \end{array} \right\} \rightarrow \text{Loose LB}$

$\left. \begin{array}{l} O(n^2) \\ O(n^3) \end{array} \right\} \rightarrow \text{Lose UB}$



Topic : Analysis of Algorithms

2. Small notations always provide bounds that are loose bounds.



(never tight bound)



Topic : Analysis of Algorithms

1. Small - Oh Notation $\rightarrow$ Loose upper Bound

$\Rightarrow f(n)$ is $O(g(n))$ iff,

for all $c > 0 \Rightarrow F(n) < c * g(n)$

When ever $n \geq n_0$

for some $n_0 \geq 0$



Topic : Analysis of Algorithms

E.g.2. Diff. between $\overset{\text{Big}}{O}$ and $\overset{\text{Small}}{o}$.

$$n < c * n$$

$$n = o(n)?$$

$$\text{if } c = 1, n < 1 * n \quad \times$$

Hence, $n \neq O(n)$

1. $f(n) = n^2 + n + 5$

$O(n^2) \rightarrow$ Tight Bound

$$n \neq o(n)$$

$$\underline{n = O(n)?} \quad \checkmark$$

~~$$n \leq 3 * n = O(n)$$~~

~~$$n = O(n^2)$$~~

~~$$n = O(n^2)$$~~

$$F(n) = O(n^3) \quad \checkmark$$

$$= O(n^2) \quad \checkmark$$

$$= O(n^2) \quad \times$$

$$= O(n^3) \quad \checkmark$$

$$= O(n\sqrt{n}) \quad \times$$

$n^2 \rightarrow$ Tight
 $\downarrow$
 $\begin{pmatrix} n^3 \\ n^4 \end{pmatrix} \rightarrow$ loose



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$$n\sqrt{n} \quad n^2$$

$$n^1 \times n^{0.5} \quad n^2$$

$$n^{(1+0.5)} \quad n^2$$

$$n^{1.5} < n^2 \quad \checkmark$$

$$\begin{array}{cc} n\sqrt{n} & n^2 \\ \cancel{n} * \sqrt{n} & \cancel{n} * n \\ \hline \sqrt{n} & n \end{array}$$

$$\begin{array}{cc} n \log n & n^2 \\ \cancel{n} \log n & \cancel{n} * n \\ \log n < & n \quad \checkmark \end{array}$$



Topic : Analysis of Algorithms

2. Small Omega (ω) Notation $\rightarrow$ Loose lower Bound

$\Rightarrow f(n)$ is $\omega(g(n))$ iff,

For all $c > 0$, $f(n) > c * g(n)$, when $n \geq n_0$

~~n_0~~ ~~$>$~~ ~~There sold~~



Topic : Analysis of Algorithms

Eg.3. Big Omega(Ω) vs small Omega (ω)

$$\boxed{f(n)=n} \Rightarrow f(n)=\Omega(n) \checkmark$$

$$=\Omega(\sqrt{n}) \checkmark$$

$$=\Omega(n^2) \times$$

$$=\omega(n) \times$$

$$=\omega(\sqrt{n}) \checkmark$$

$$=\omega(1) \checkmark$$



$$f(n)=n$$

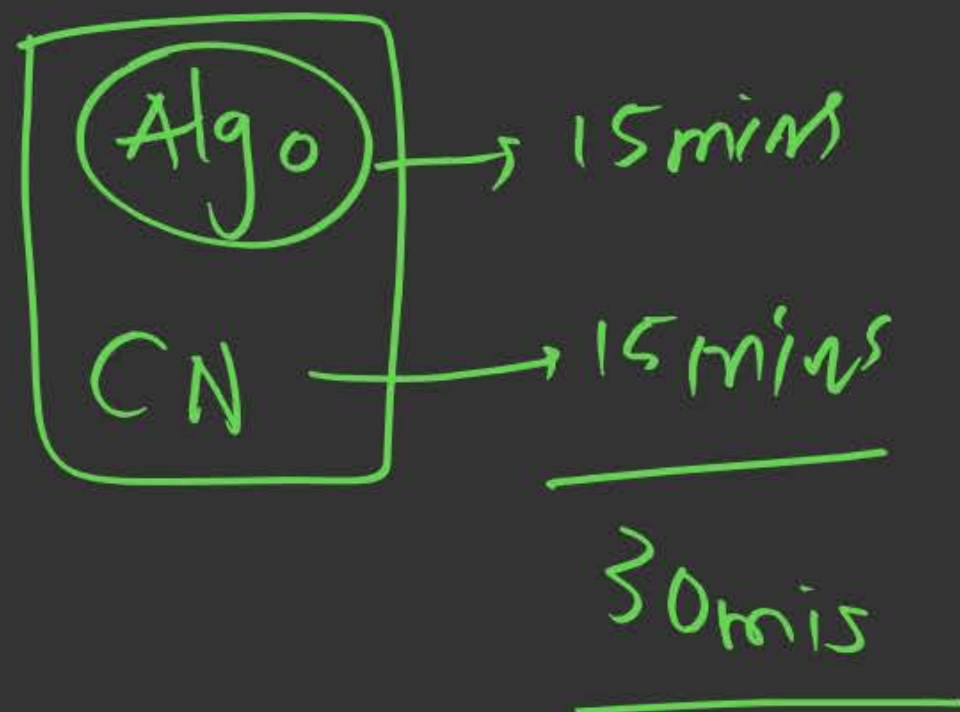
$$n \geq c * n \rightarrow \Omega(n)$$

Tight LB

$$n > c * n \times$$

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Algo
CN



TOC
CD
Apti
DL
C

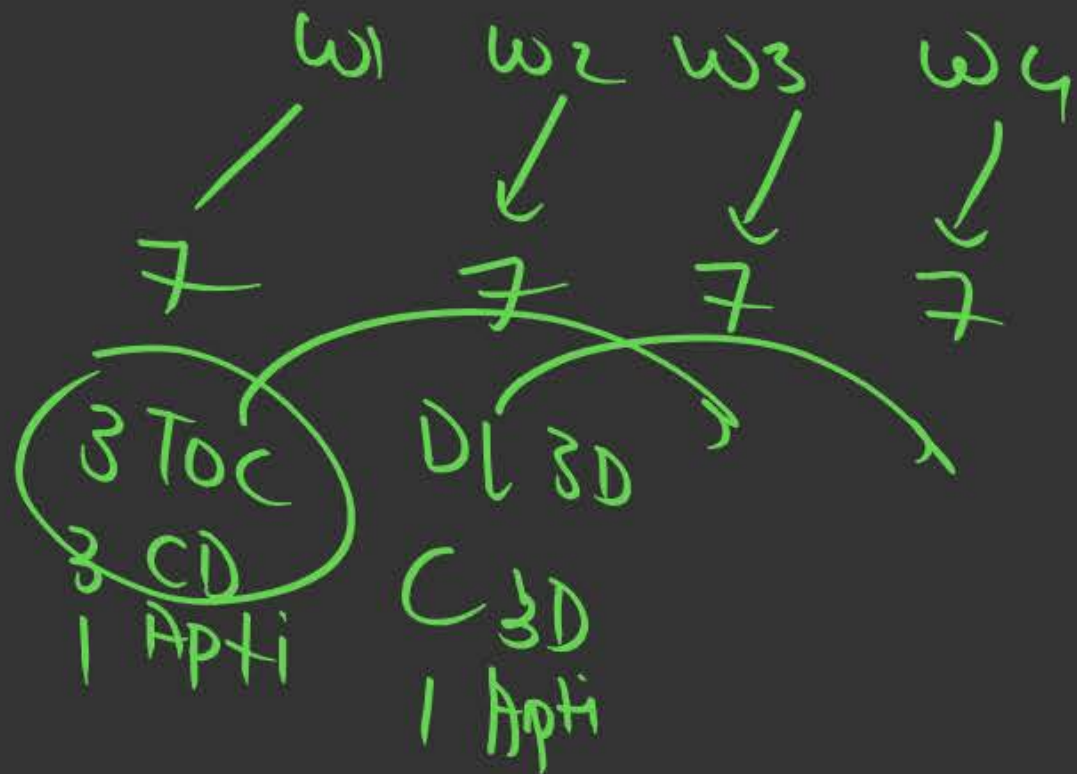
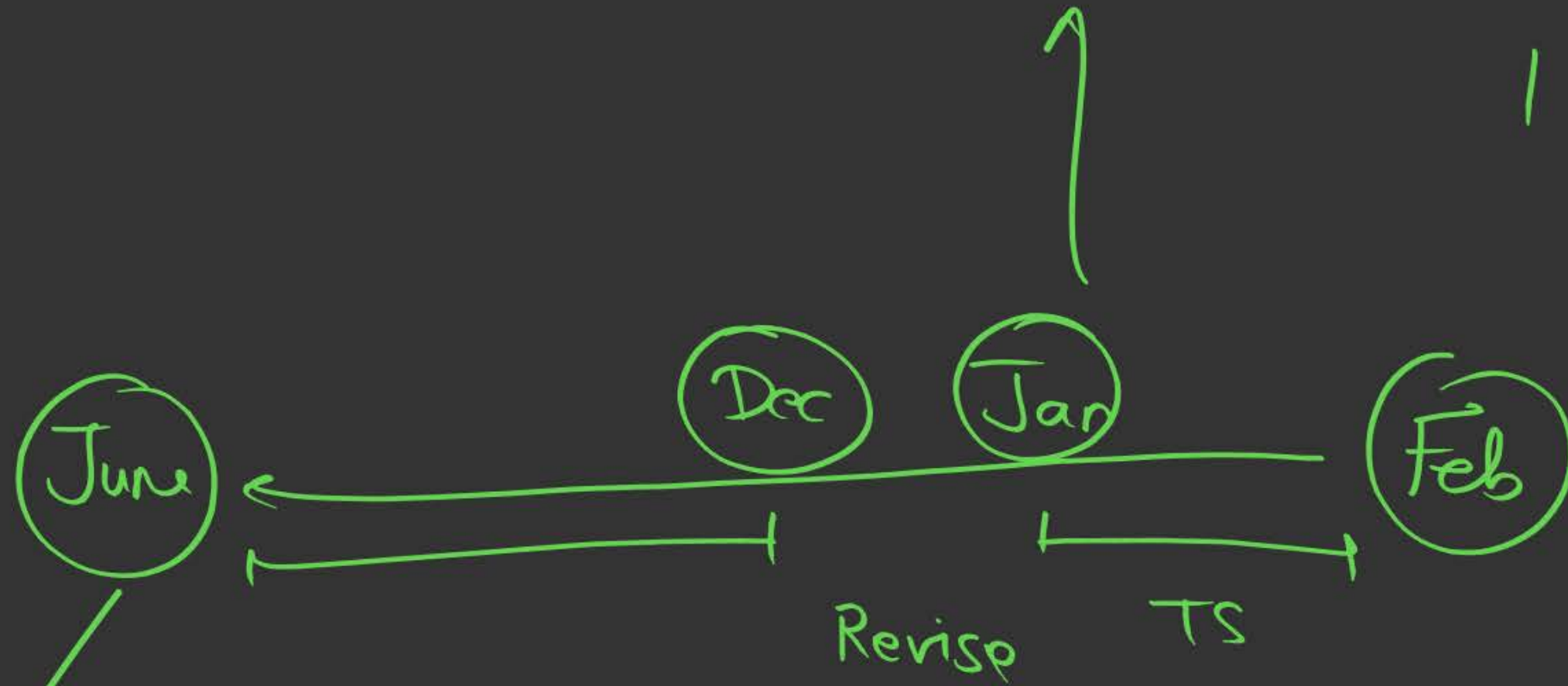
PYQ
DPP

2.5 hrs

Revise poor Subjects

~~MTWTFSS~~

1 subj $\rightarrow$ 2 days



AIR 60

77.33/100

C/g +
placement

Practice Questions:

T/F



Topic : Analysis of Algorithms

Imp. Practice Questions

#Q. $2(2^n) = O(2^n)$

$$2 \times 2^n = O(2^n) ? \quad \underline{\underline{\text{True}}}$$

$$2^n \stackrel{A}{=} 2^n$$

$$(Q) \quad 2^{(2n)} = O(2^n) \rightarrow \text{False}$$

$$(a^m)^n = (a^n)^m = a^{m \times n}$$

$$\begin{aligned} & 2^{2n} \\ &= (2^2)^n \\ &= 4^n > 2^n \end{aligned}$$

$$2^{2n} \neq O(2^n)$$

$$2^{(2n)} > 2^{(n)}$$

Take log

$$\log_2(2^{2n}) \quad \log_2(2^n)$$

$$2n > n$$



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Imp. Practice Questions

#Q. If $0 < a < b$ then $n^a = O(n^b)$

True

$$\underline{a < b}$$

$$\underline{n^a < n^b}$$



Topic : Analysis of Algorithms

Imp. Practice Questions

#Q. $2^{(n^2)} = O(n!)$ $\longrightarrow$ **False**

$$2^{n^2} > n^n$$

Take log

$$\log_2(2^{n^2}) > \log_2(n^n)$$

$$n^2 > n \log(n)$$

$$n^2 \text{ vs } n \log n$$

$$n \times n > n \times \log n$$

$$n > \log n$$

#Q. Consider the following functions from positive integers to real number:

$$10, \sqrt{n}, n \log_2 n, \frac{100}{n}$$

The correct arrangement of the above functions in increasing order of asymptotic complexity is:

64%

A

$$\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$$

B

$$10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$$

C

$$\frac{100}{n}, 10, \log_2 n, \sqrt{n}, n$$

D

$$\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$$

10

$\sqrt{n}$

n

$\log n$

$$\boxed{\frac{100}{n}}$$

→ decreasing

~~~~~

$$n > \sqrt{n} > \log n > 10 > \frac{100}{n}$$

$$\text{inc: } \frac{100}{n} < 10 < \log n < \sqrt{n} < n$$

---

Proof

$n = 64$

$$\sqrt{n} \quad \log_2 n$$

$$\sqrt{64} \quad \log_2 64$$

$$8 > 6$$

$$\sqrt{n} = n^{1/2} > \log n$$

$$\log(n^{1/2}) \quad \log(\log n)$$

$$\frac{1}{2} \log n$$

$$\log \log n$$

$$\log n \rightarrow n$$

$$n^{1/2} > \log(n)$$





## Topic : General Properties of Big Oh Notation

Let  $d(n)$ ,  $e(n)$ ,  $f(n)$ , and  $g(n)$  be functions mapping nonnegative integers to nonnegative reals. Then

1. If  $d(n)$  is  $O(f(n))$ , then  $a \times d(n)$  is  $O(f(n))$ , for any constant  $a > 0$
2. If  $d(n)$  is  $O(f(n))$  and  $e(n)$  is  $O(g(n))$ , then  $d(n) + e(n)$  is  $O(f(n) + g(n))$ .
3. If  $d(n)$  is  $O(f(n))$  and  $e(n)$  is  $O(g(n))$ , then  $d(n) \times e(n)$  is  $O(f(n) \times g(n))$
4. If  $d(n)$  is  $O(f(n))$  and  $f(n)$  is  $O(g(n))$ , then  $d(n)$  is  $O(g(n))$ . Transitivity
5. If  $f(n)$  is a polynomial of degree  $d$  (that is,  $f(n) = (a_0 + a_1n + \dots + a_d n^d)$ ) then  $f(n)$  is  $O(n^d)$ .
6.  $n^x$  is  $O(a^n)$  for any fixed  $x > 0$  and  $a > 1$
7.  $\log(n^x)$  is  $O(\log n)$  for any fixed  $x > 0$
8.  $\log^x n$  is  $O(n^y)$  for any fixed constants  $x > 0$  and  $y > 0$



## Topic : General Properties of Big Oh Notation

1. If  $d(n) = O(f(n))$ ,

then  $a * d(n) = \underline{O(f(n))}$ ,  $a > 0$

e.g.:  $d(n) = n^2 \rightarrow O(n^2)$

$a = 10$

$a * d(n) = 10 * n^2 \rightarrow O(n^2)$





## Topic : General Properties of Big Oh Notation

2.  $d(n) = 5n + 2 \rightarrow O(n)$

$e(n) = 10n^2 + 7n + 9 \rightarrow O(n^2)$

$d(n) + e(n) = (10n^2 + 7n + 9) + (5n + 2)$

$= (10n^2 + 12n + 11)$

$= O(n^2)$

$\Rightarrow O(f(n) + g(n))$

$\Rightarrow O(n + n^2)$

$\Rightarrow O(n^2)$

**Shortcut:**  $O(f(n) + g(n)) = O(\max(f(n), g(n)))$







## Topic : General Properties of Big Oh Notation

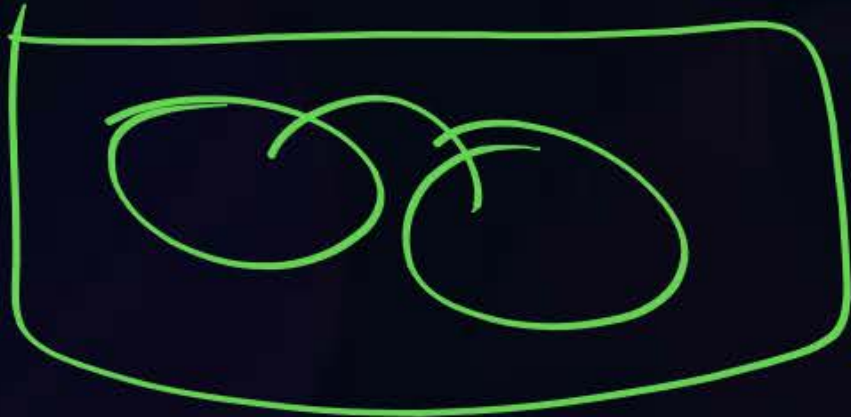
3.  $d(n) = 5n \rightarrow O(n)$

$e(n) = 10n^2 + 2 \rightarrow O(n^2)$

$$d(n) * e(n) = (10n^2 + 2) * (5n)$$

$$= (50n^3 + 10n)$$

$$= \underline{\underline{O(n^3)}}$$



$$\Rightarrow O(f(n) * g(n))$$

$$\Rightarrow O(n * n^2)$$

$$\Rightarrow \underline{\underline{O(n^3)}}$$



## Topic : General Properties of Big Oh Notation

4.  $d(n) = O(f(n))$  and  $f(n) = O(g(n)) \Rightarrow d(n) = O(g(n))$

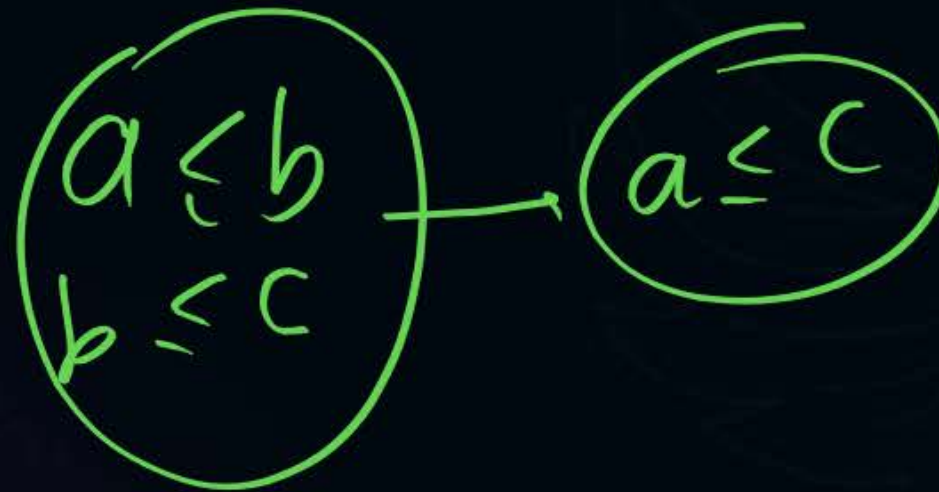
e.g.:

$$d(n) = O(n^3)$$

and  $f(n) = O(n^4)$

then  $d(n) = O(n^4)$

Transitivity







## Topic : General Properties of Big Oh Notation

5.  $f(n) = 5 + 10n + 15n^3 + 7n^4$

$$f(n) = O(n^4)$$



## Topic : General Properties of Big Oh Notation

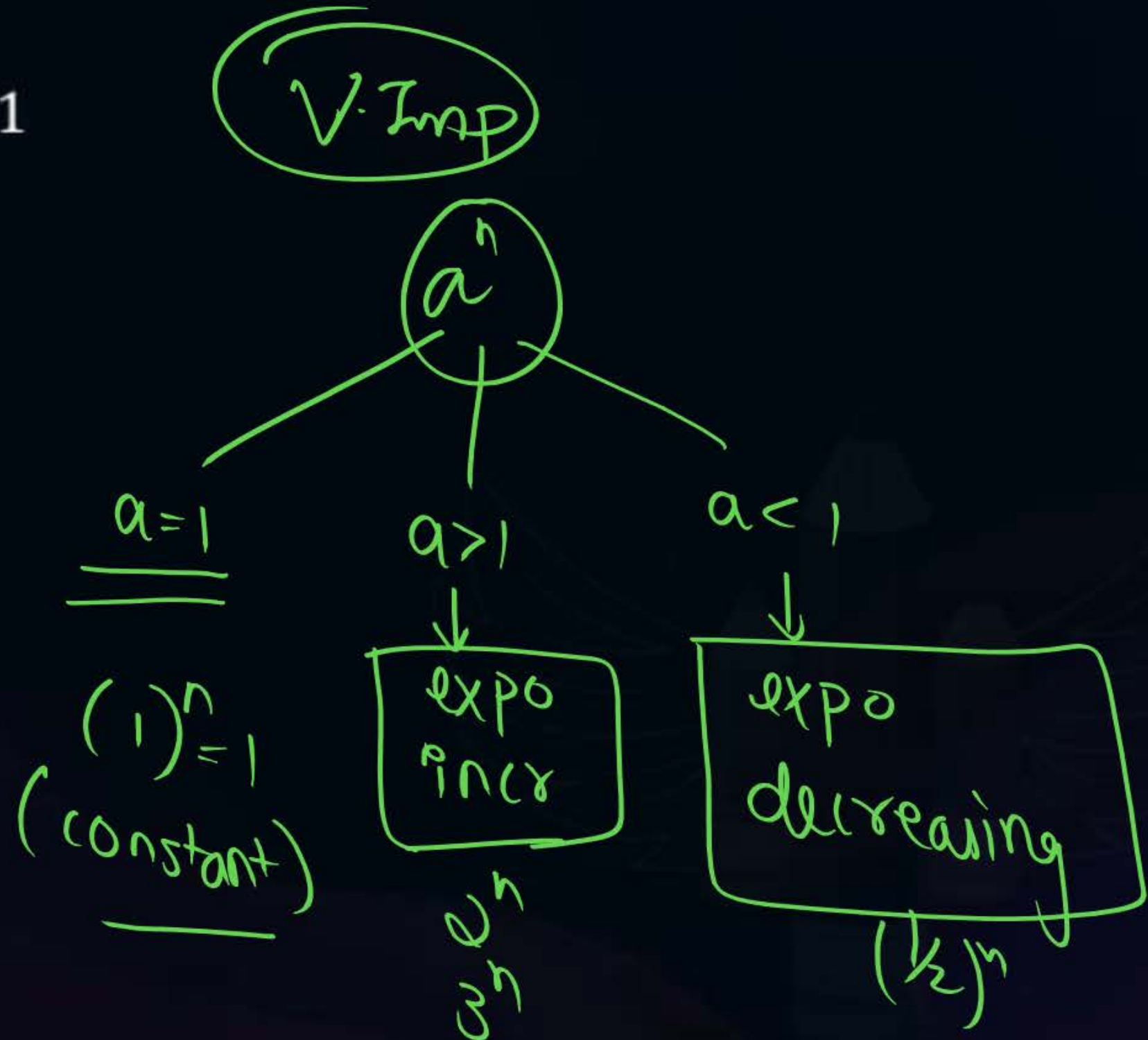
6.  $n^x = O(a^n)$ , for any  $x > 0$  &  $a > 1$

$n^x, x > 0 \rightarrow$  Polynomial

$a^n, a > 1 \rightarrow$  Exponential

Poly < Expo

Poly =  $O(\text{Expo})$









## Topic : General Properties of Big Oh Notation

7.  $f(n) = \log(n^x) = x * \log(n) = O(\log n)$

$$f(n) = \log(n^3) = 3 * \log(n) \rightarrow O(\log n)$$

$$f(n) = \log(n^6) = 6 * \log(n) \rightarrow O(\log n)$$

$$f(n) = \log(n^{10}) = 10 * \log(n) \rightarrow O(\log n)$$

$$f(n) = \log(n^{100}) = 100 * \log(n) \rightarrow O(\log n)$$

$$n^2 < n^{100}$$

$$\underline{O(\log n)}$$





## Topic : General Properties of Big Oh Notation

8.  $(\log n)^x$  or  $\log^x(n)$  =  $O(n^y)$ ,  $x > 0, y > 0$

Logarithmic < Polynomial

$\log = O(\text{poly})$

$$\begin{array}{l|l} (\log n)^{100} < n^2 & \log^2(n) = (\log(n))^2 \\ & \neq \log(n^2) \end{array}$$



$$[\log(n)]^{100} \quad n^2$$

Take log both sides

$$\log((\log n)^{100}) \quad \log(n^2)$$

$$100 * \log(\log n) \quad 2 * \log(n)$$

Let  $\boxed{\log n = x}$

$$\underline{100 * \log(n)} < \underline{2 * x}$$



# Summary



→ Asymptotic notations

→ Properties

→ Practice



**THANK - YOU**